



Understanding the Weather



Do you ever think how do weather changes affect our daily life? What causes rain, storms and different seasons?

Let's explore these types of questions and discover the answers in this chapter.



Learning Outcomes

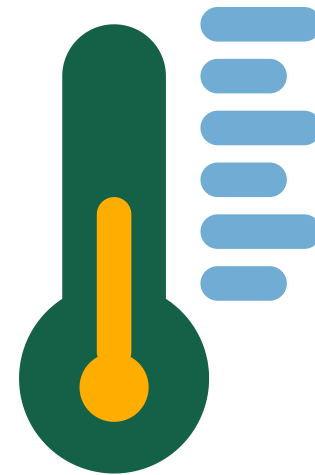
By the end of the lesson, you'll be able to:

1



Understand the meaning of weather and its elements

2



Understand how meteorologists predict weather

3



Explain how weather prediction helps people in daily life



Before we understand the elements and instruments of weather, we must first know the difference between Atmosphere, weather and climate.



Atmosphere	Weather	Climate
The atmosphere is a thick blanket of air that surrounds the Earth. • It protects Earth from harmful rays of the Sun. • It helps maintain temperature on Earth.	Weather is a Day-to-day condition of the atmosphere. • It is the condition of the atmosphere at a particular place and time. • It may be sunny, rainy, cloudy, windy, hot, or cold.	Climate is the average weather condition of a place over a long period of time. • It includes temperature, rainfall, humidity and wind patterns of a region. • Climate does not change daily like weather.



ATMOSPHERE

The atmosphere is the layer of gases surrounding Earth, held in place by gravity. This protective envelope shields life from harmful radiation and regulates temperature.

It is divided into different layers based on temperature and height above the Earth. The lowest layer, called the **troposphere**, is where humans, animals, and plants live and where weather changes occur. It extends to a height of 6 to 18 kilometres from the ground; it is **less thick at the poles** and **thicker in the tropical zone**.

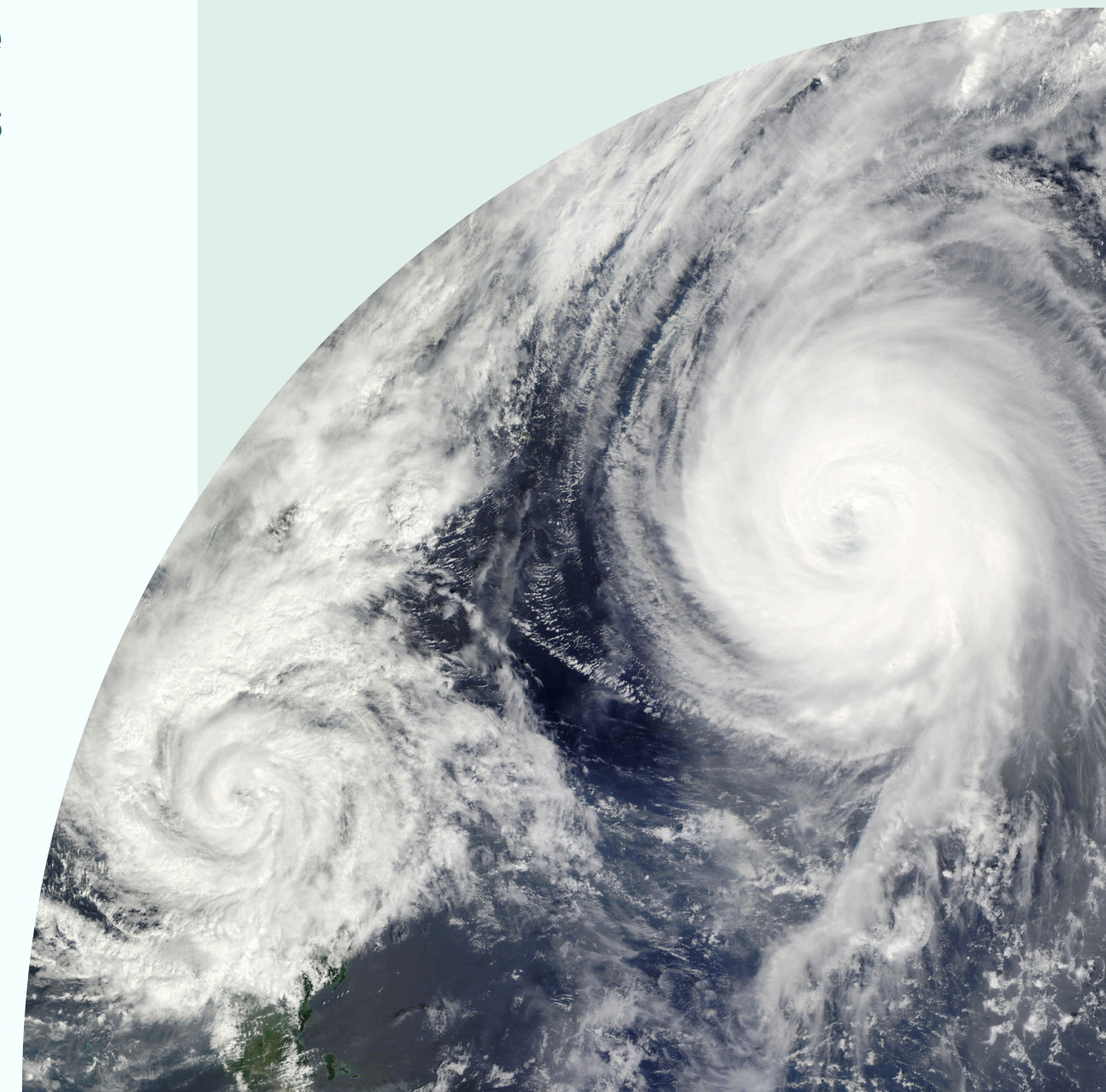
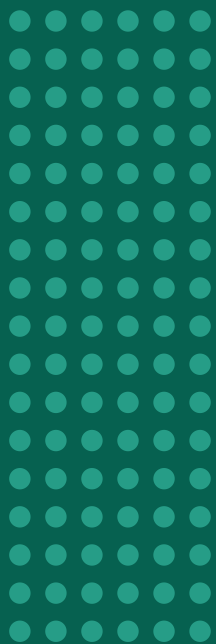
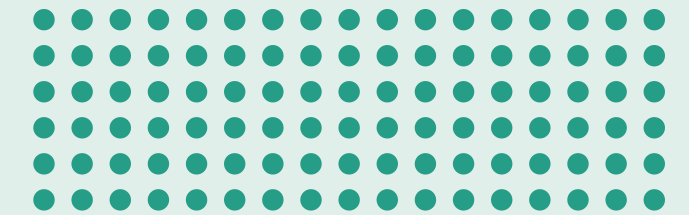


Weather and its elements

The elements of weather are the basic factors that describe what is happening in the atmosphere at a given time and place or we can say, these are the different conditions of the atmosphere that help us understand the weather of a place.

The elements of weather are:

- Temperature
- Precipitation
- Atmospheric Pressure
- Wind
- Humidity



Elements of Weather

Temperature



Level of hotness or coldness of the air at a particular place and time.

Precipitation



Any form of water (rain, snow, sleet or hail) that falls from clouds to the surface of the Earth.

Atmospheric Pressure



Force exerted by the air on the surface of the Earth.

Wind

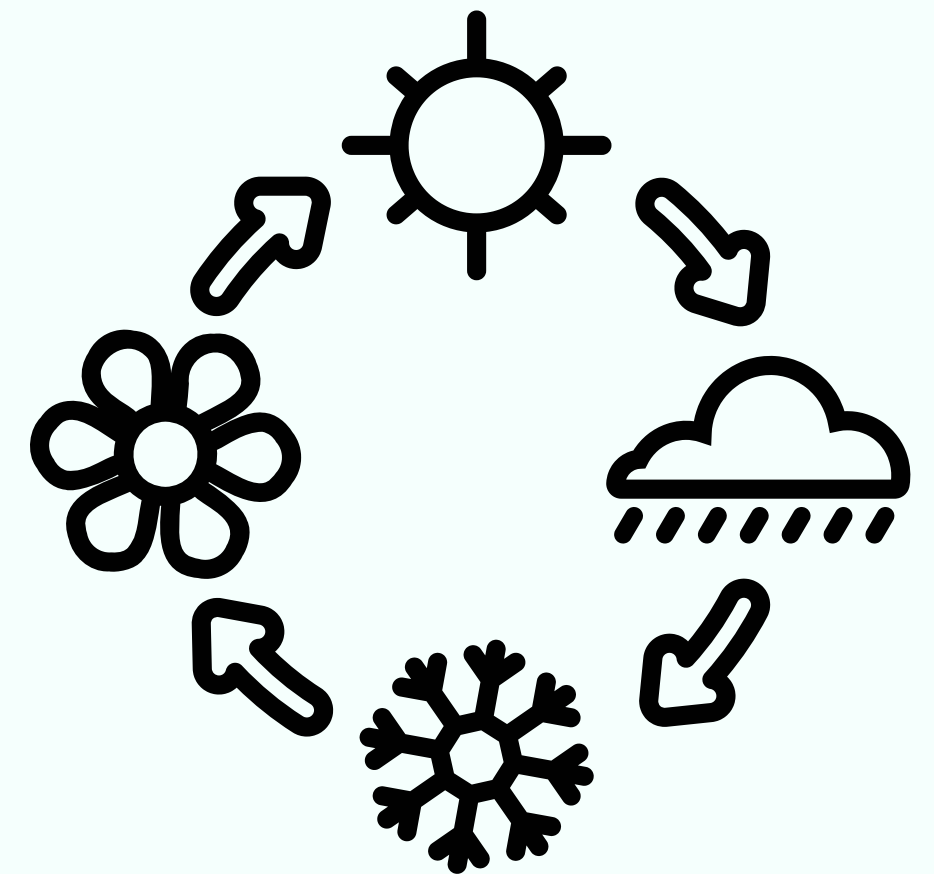
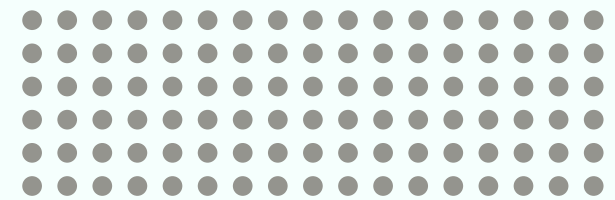


Movement of air from one place to another

Humidity



It is the amount of water vapour present in the air.



QUIZ TIME



Which of the following statements is NOT true regarding the troposphere?

A. It is the lowest layer of the atmosphere.

B. All weather activities takes place in this layer.

C. It is thicker at poles and thinner at tropical zone.

